

Exponential Growth & Decay Applications (3.5)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: 6.93

_____ (**Calculator**) \$500 is invested at 6% compounded continuously for 5 years. Find the balance to the nearest cent.

ANSWER: \$1,157.63

_____ (**Calculator**) A culture is modeled by $y = 100e^{0.2t}$. Find the amount at $t = 5$, to the nearest hundredth.

ANSWER: 20 years

_____ A substance has a half-life of 6 years. How many years until only $\frac{1}{8}$ remains?

ANSWER: 17.33 years

_____ A 80 mg sample has a half-life of 4 hours. How many mg remain after 12 hours?

ANSWER: \$674.93

_____ A population doubles every 10 years. How many years until it is 4 times its original size?

ANSWER: 10 mg

_____ **(Calculator)** A sample decays by $y = 50e^{-0.1t}$. Find t when $y = 25$, to the nearest hundredth.

ANSWER: 18 years

_____ **(Calculator)** \$1,000 is invested at 5% compounded annually for 3 years. Find the balance to the nearest cent.

ANSWER: 271.83

_____ **(Calculator)** An account grows continuously at 4%. Find the doubling time in years, to the nearest hundredth.